HOW WELL DO NEURAL AUDIO CODECS PRESERVE PERCEPTUAL INFORMATION? A LAYER-WISE ANALYSIS OF ENCODEC AND DAC

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ABSTRACT

Neural audio codecs such as EnCodec and DAC were designed for efficient audio compression, yet they are increasingly used as the foundation for creative audio applications including music generation and latent granular resynthesis. It's not very clear if their residual vector quantization (RVQ) structure preserves perceptual information in a way that supports creative use. We probe each codebook layer of both codecs using acoustic feature detectors for pitch, spectral centroid, flatness, MFCCs, and chroma, across five audio types. We find that EnCodec concentrates perceptual information in its earliest codebook layers while DAC distributes it more evenly across all 32 layers.

1. BACKGROUND AND RELATED WORK

The last few years, neural audio codecs have been a topic of discussion and their use has escaped their original purpose. Encodec [1] and DAC [2], two of the most known codecs, have been used creatively for different models: audio manipulation and music generation. Since their original architecture didn't have that in mind, it is obvious to ask: Is there any perceptual meaningfulness in the way they encode the audio information across different codebook layers?

MusicGen uses EnCodec tokens as its internal representation of music, generating new compositions by predicting sequences of codebook indices. Tokui's latent granular resynthesis paper repurposes EnCodec's quantized latents for sound design, matching grains of one audio corpus to the temporal structure of another.

VampNet uses DAC tokens for music generation through masked acoustic token modelling: masking portions of a DAC token sequence and training a model to predict the missing tokens, enabling creative operations like music continuation, variation, and style transfer. HummingLM uses DAC's compressed audio representation to transform hummed melodies into professional instrumental performances. Also, MusicGen's authors tested DAC as a replacement for EnCodec but ultimately preferred EnCodec. In each of these applications, both codecs are being asked to do something their designers never intended — to serve as musically and perceptually meaningful latent spaces for creative manipulation

Each codebook layer encodes residual error from the previous one, meaning information is distributed across layers in a way that is not perceptually interpretable by design. Understanding what perceptual information lives where requires external probing tools.

DAC was designed to address several known limitations of EnCodec. While EnCodec was optimised for low bitrate transmission at the cost of audio quality, DAC prioritised higher fidelity reconstruction for professional audio and generative model use. One important technical contribution was addressing codebook collapse: a problem where large fractions of each codebook's 1024 entries remain unused during encoding, "wasting" the codec's representational capacity. DAC introduced periodic random codebook resets during training, forcing all entries to remain active and resulting in substantially higher codebook utilization.

In terms of architecture, both EnCodec and DAC support up to 32 codebooks at 24kHz, however they differ in how codebook count is exposed to the user. EnCodec treats the number of active codebooks as a variable bandwidth parameter: using 8 codebooks at 6kbps or 32 codebooks at 24kbps. DAC uses all 32 codebooks as its standard operating configuration. In this study we evaluate EnCodec at both settings. 8 codebooks is its most common creative use configuration and 32 codebooks is encodec's full capacity (equivalent to DAC), allowing us to separate the effect of codebook count from the effect of codec architecture.

Both codecs were explored using five acoustic feature detectors: pitch, spectral centroid, spectral flatness, MFCCs, and chroma, applied to cumulative reconstructions using increasing numbers of codebook layers, across five audio types spanning a range of acoustic complexity. Our goal is not to evaluate the codecs as compression tools, but to understand their internal perceptual structure and what it means for the growing ecosystem of creative applications that depend on them.

2. METHODOLOGY

For the experiment we used 5 audio files with different structure, extracted from freesound, to explore the codecs' behavior in each case. The audios were:

- speech: alternates between voiced vowels with clear pitch and unvoiced consonants with no fundamental frequency

- singing voice: sustained pitched notes with vibrato and time-varying vowel formants, very complex for timbre

- flute: nearly pure tone with a clean harmonic series, minimal noise, and highly repetitive frames, acoustically more simple

- saxophone: combines a rich harmonic series with breath noise from the reed mechanism, making it simultaneously pitched and noisy.

- amen percussion: an unpitched drum loop with sharp transient attacks and fast rhythmic variation, producing the highest codebook utilization.

All clips were resampled to 24kHz mono using Audacity. Both codecs were loaded using their official pretrained weights. EnCodec was loaded via the HuggingFace Transformers library and DAC was installed via the official descript-audio-codec Python package and loaded using its built-in download utility (model type="24khz"), using all 32 codebooks at full capacity.

For each codec and audio type, audio was reconstructed using progressively more codebook layers, from Q1 alone up to the full set. For EnCodec@8 all 8 levels were decoded, while for EnCodec@32 and DAC@32 a sparse set of 12 levels was used, sampling at the early layers where most perceptual change occurs. The primary comparison was based on the creative usage: EnCodec was used at 8 codebooks by MusicGen, and DAC at 32 codebooks by VampNet. To also have an "equal" architectural comparison, EnCodec at 32 codebooks was also evaluated, with results normalised to percentage of total codebooks used on the x axis. All models were used in inference mode with pretrained weights and no fine-tuning.

The goal was to see how both codecs' different codebooks behave regarding different downstream tasks. Specifically, we extracted for each codebook layer the following features through Librosa library.

2.1 Acoustic features extraction

- **Pitch (F0)** was estimated using the probabilistic YIN algorithm (pYIN), searching for fundamental frequencies between C2 (65 Hz) and C7 (2093 Hz). Frames with no detected pitch were returned as `NaN` and excluded from correlation computation. This feature is only meaningful for voiced or tonal frames. To improve reliability, frames with voicing probability below 0.5 were additionally excluded, and a median filter was applied to remove octave jump artifacts. This feature is only meaningful for voiced or tonal frames.

- **Spectral Centroid** measures the centre of mass of the frequency spectrum, serving as a proxy for perceived brightness. It is computed on every frame regardless of voiced/unvoiced status.

- **Spectral Flatness** measures the ratio of the geometric to arithmetic mean of the power spectrum, quantifying how tonal versus "noisy" a frame is. A value near 0 indicates a pure tone and a value near 1 indicates white noise.

- **MFCCs** (Mel-Frequency Cepstral Coefficients, 13 coefficients) capture the spectral envelope shape of the signal, encoding timbral texture independently of pitch.

- **Chroma (CQT-based)** measures the distribution of energy across the 12 pitch classes (C through B) using a Constant-Q Transform, capturing harmonic content and musical note distribution regardless of octave.

2.2 Evaluation

For each feature, codec, audio type, and codebook level, the Pearson correlation coefficient was computed between the feature extracted from the partial reconstruction and the feature extracted from the full reconstruction (using all codebooks). This creates a score in $[-1, 1]$, where 1.0 indicates perfect preservation, 0.0 indicates no relationship, and negative values indicate anticorrelation.

For one-dimensional features (pitch, centroid, flatness), the correlation is computed directly after removing `NaN` frames. For multi-dimensional features (MFCC, chroma), the correlation is computed per dimension and averaged:

$$r = \frac{1}{D} \sum_{d=1}^{D} \operatorname{corr}\Big(\hat{f}_d,\ f_d^{\mathrm{ref}}\Big) \tag{1}$$

where $\hat{f}_d$ is the d-th dimension of the feature extracted from the partial reconstruction and f_d^{ref} is the corresponding dimension from the full reconstruction. Frames where either signal has zero variance are excluded to avoid undefined correlations.

In addition, since DAC claims that it solves Encodec's capacity problem (it uses a small percentage of the codebooks' capacity), we computed the percentage of "words" used for each audio. The results were plotted, and for the most interesting shifts in the results audios were also presented.

3. RESULTS

3.1 Pitch

Pitch is one of the first features to be encoded in EnCodec. For tonal audio types (singing, flute, saxophone), Q1 alone captures pitch with almost perfect correlation (0.996-1.000) across both EnCodec configurations. This makes sense if we think that pitch is a coarse low-frequency property. Speech and amen return NaN throughout all codecs and levels, reflecting that neither contains sufficient stable pitched content for reliable measurement: an expected result given speech's rapid voiced/unvoiced alternation and the amen break's percussive character. DAC captures almost no pitch at Q1 for any audio type the detector returns NaN for all tonal sounds, confirming that DAC's distributed architecture does not preserve pitch information in its earliest layer.

3.2 Spectral Centroid

Spectral centroid measures the brightness of a sound. EnCodec preserves this well at Q1 for most audio types (0.723-0.976), with singing being consistently the hardest (0.723-0.775). DAC shows more variable centroid preservation: flute is reasonably well preserved (0.897) while speech is near zero or negative (-0.099). The negative value for DAC speech centroid suggests that Q1 alone decodes to audio with spectral characteristics inversely related to the original, likely because the decoder is severely underconditioned with only 3.1 percent of its expected input.

3.3 Flatness

Flatness is well preserved at Q1 for all audio types, better than MFCC and chroma. This makes sense because flatness is a coarse spectral property. Singing has the lowest flatness correlation (0.813) probably because singing has the most time-varying spectral character (vowels have very low flatness), consonants and breaths have high flatness, and Q1 struggles to capture these rapid transitions precisely.

One interesting thing is that DAC Q1 preserves flatness better than pitch or chroma for most audio types. Flute flatness at DAC Q1 is 0.924 despite pitch being -0.023. This reveals that DAC's first codebook encodes the general spectral character of a sound without encoding its specific harmonic structure.

3.4 MFCC

MFCC captures timbral texture and shows the most variation across audio types and codecs, making it the most interesting feature in this study. EnCodec Q1 MFCC ranges from 0.389 (singing) to 0.721 (speech), showing that singing's complex time-varying timbre requires more codebook layers to reconstruct. DAC@32 is consistently lower than both EnCodec configurations across four out of five audio types, confirming that timbral texture is poorly preserved in DAC's earliest layer. The single exception is singing, where DAC@32 Q1 (0.479) marginally outperforms both EnCodec@8 (0.389) and EnCodec@32 (0.349). However, this shouldn't be interpreted naively: MFCC correlation measures whether the spectral envelope varies in the same pattern as the full reconstruction, not absolute perceptual quality. DAC@32 singing Q1 produces no detectable pitch (NaN) and sounds clearly degraded in the audio results, suggesting that the higher MFCC score reflects a loose spectral envelope correlation rather than genuine timbral preservation.

3.5 Chroma

Chroma measures pitch class distribution. It directly captures harmonic content and is the most affected by DAC's distributed architecture. EnCodec Q1 chroma values are high for tonal sounds (flute 0.888, singing 0.857) and low for percussive sounds (amen 0.485). DAC Q1 chroma is near zero for almost all audio types (0.008-0.143), with singing being the only exception (0.267 for pitch, 0.039 for chroma). This confirms that harmonic structure is the last perceptual property to emerge in DAC's layer hierarchy — requiring many codebooks before the harmonic relationships between frequencies become coherent.

Feature	Audio	ENC@8	ENC@32	DAC@32
Pitch	Speech	NaN	NaN	NaN
	Singing	**0.996**	**0.996**	NaN
	Flute	**1.000**	**1.000**	NaN
	Saxophone	**1.000**	**1.000**	0.000
	Amen	**NaN**	NaN	NaN
MFCC	Speech	**0.718**	0.721	0.527
	Singing	0.389	0.349	**0.479**
	Flute	**0.716**	0.708	0.424
	Saxophone	**0.505**	0.473	0.305
	Amen	**0.624**	0.608	0.392
Chroma	Speech	**0.616**	0.565	0.143
	Singing	**0.857**	0.847	0.039
	Flute	**0.888**	0.867	0.072
	Saxophone	0.755	**0.764**	0.016
	Amen	**0.485**	0.457	0.008
Flatness	Speech	0.870	**0.876**	0.141
	Singing	0.813	**0.857**	0.705
	Flute	**0.947**	0.931	0.924
	Saxophone	**0.938**	0.900	0.304
	Amen	**0.927**	0.921	0.605
Centroid	Speech	**0.976**	0.973	−0.099
	Singing	0.775	0.723	**0.685**
	Flute	**0.942**	**0.942**	0.897
	Saxophone	**0.906**	0.889	0.483
	Amen	**0.925**	0.914	0.837

Table 1. Pearson correlation between acoustic features extracted from the Q1-only reconstruction and the full reconstruction, for each codec and audio type. Bold indicates the highest value per row. Negative values indicate anticorrelation.

4. CONCLUSION

This study investigated whether neural audio codecs preserve perceptual information in a way that supports creative audio manipulation, according to the question: how well do downstream perceptual detectors function as codebook layers are progressively removed?

One finding was that EnCodec and DAC seem to use different information distribution strategies. EnCodec encodes perceptual content from its earliest layers. Pitch and brightness can be largely recovered from Q1 alone across all audio types, making individual layer manipulation musically meaningful. DAC distributes information gradually across all 32 layers, meaning no single layer carries enough perceptual content to support downstream detection reliably.

Comparing EnCodec at 8 and 32 codebooks showed that how much information ends up in Q1 is not fixed, but it depends on how many codebooks the codec has available. When EnCodec uses 32 codebooks instead of 8, it "spreads" the encoding more evenly across all layers, meaning each individual layer carries less. This has a practical consequence: as codecs evolve towards higher quality by adding more codebook layers, the first layer becomes less and less informative on its own.

Codebook utilization was consistently low for both codecs (8-27 percent of 1,024 entries), which probably is attributed to the nature of the audio clips used: short, relatively homogeneous sounds that do not generate enough acoustic diversity to populate the codebook richly. Longer and more varied material would likely produce higher utilization across all layers. Also, DAC's utilization improvement may be partly a consequence of its distributed architecture: fine-grained residuals are naturally more diverse than coarse approximations, activating more entries regardless of training technique.

These results verify the initial idea that these creative tools built on neural codecs are relying on properties that were never designed for encoding meaningful information. EnCodec@8 is a very accessible codec for creative layer manipulation, while DAC is better suited for applications using the full token sequence. Building codecs explicitly designed for layer-level interpretability, where early layers encode pitch, later layers encode timbre, is a very interesting challenge, with SpeechTokenizer pointing towards one possible direction for speech.

5. REFERENCES

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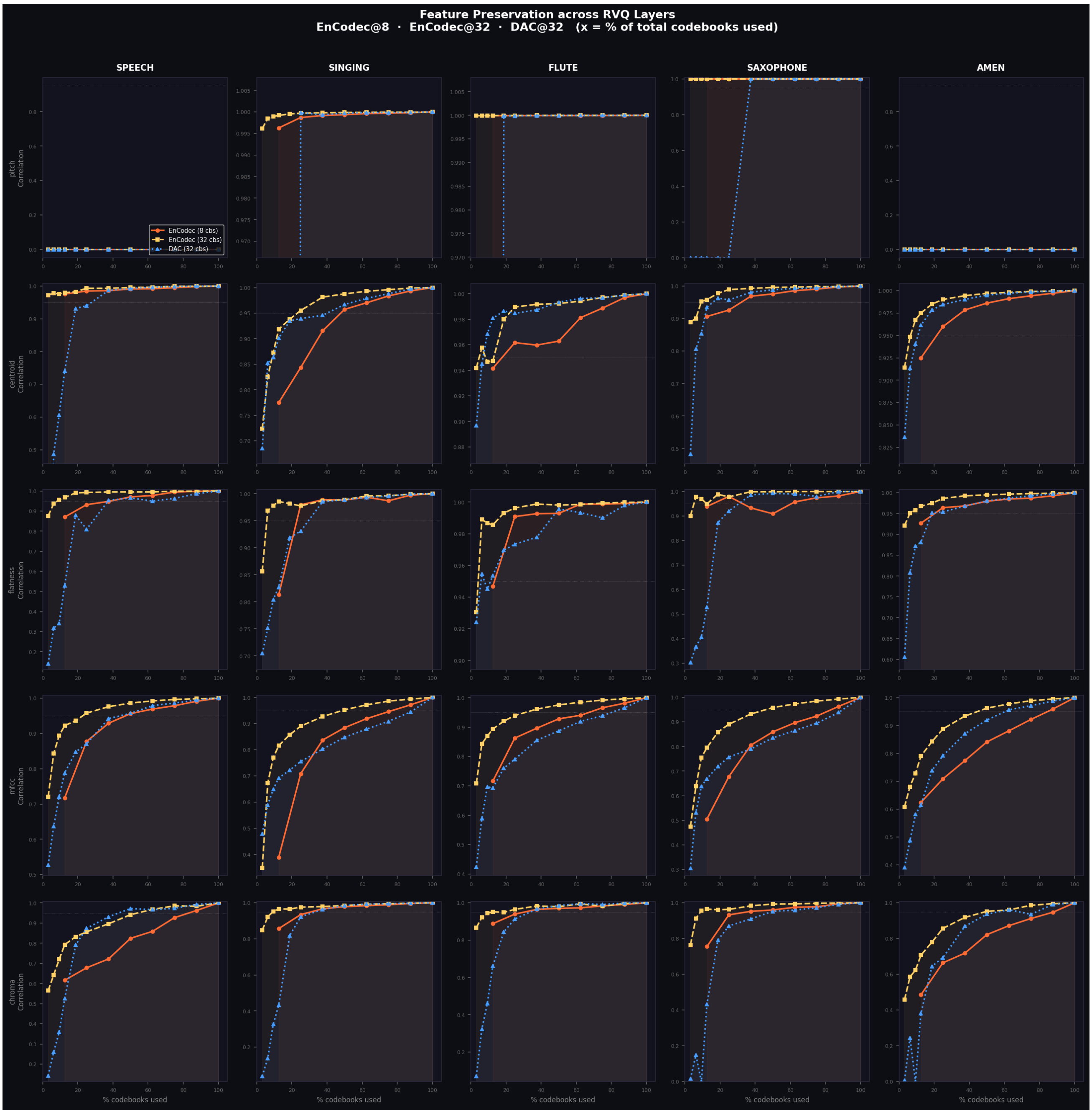

Figure 1. Plots of all the codecs' results accross all features and different codebooks levels .

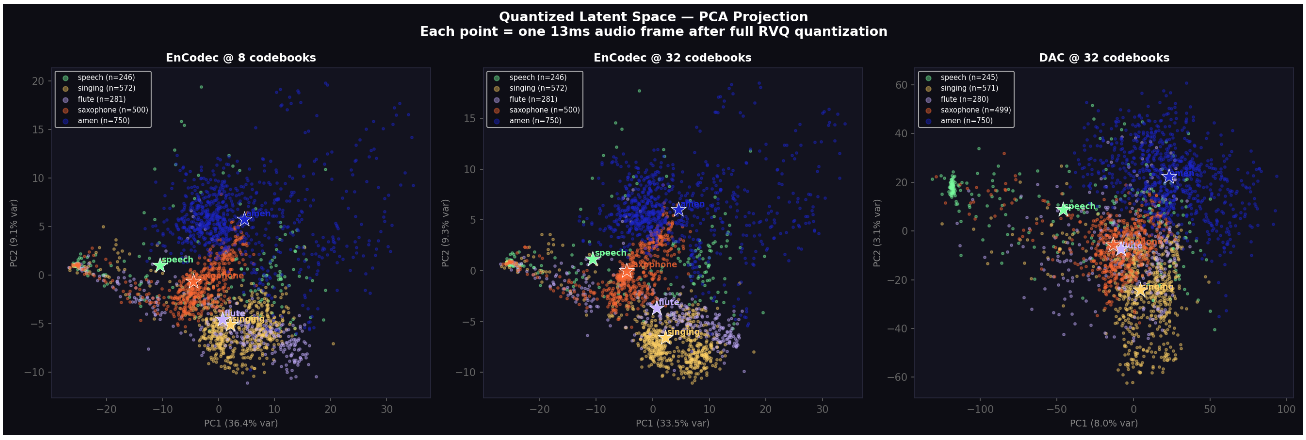

Figure 2. Visualization of quantized latent space.

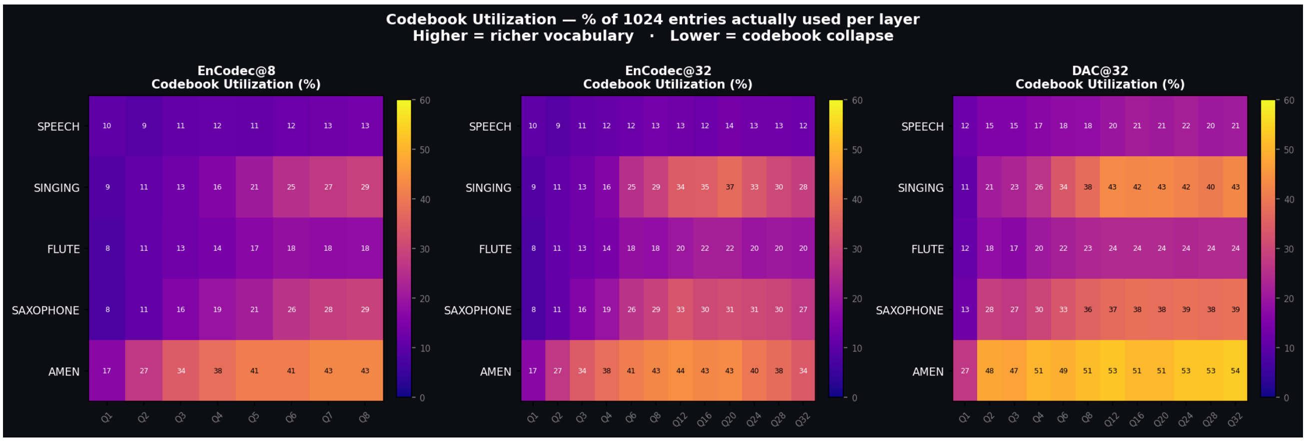

Figure 3. Heatmap of codebook utilization.